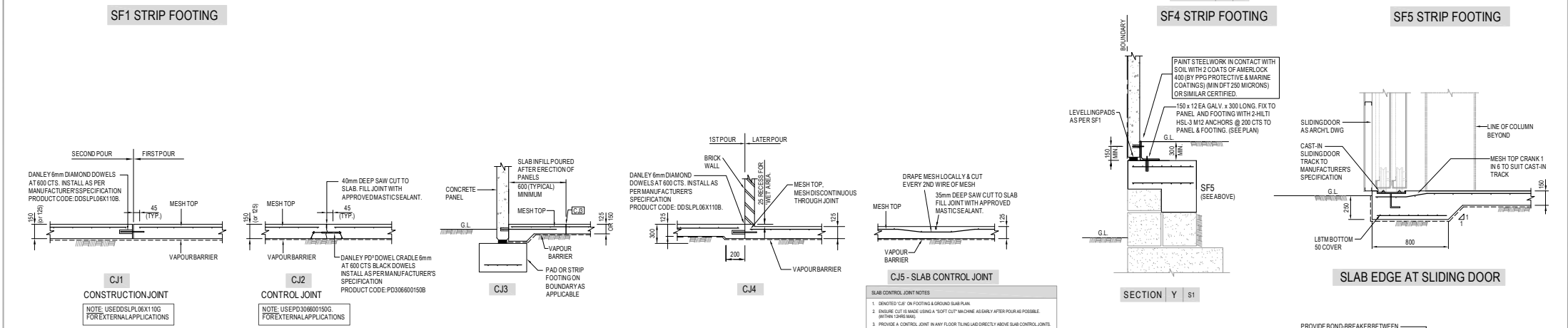
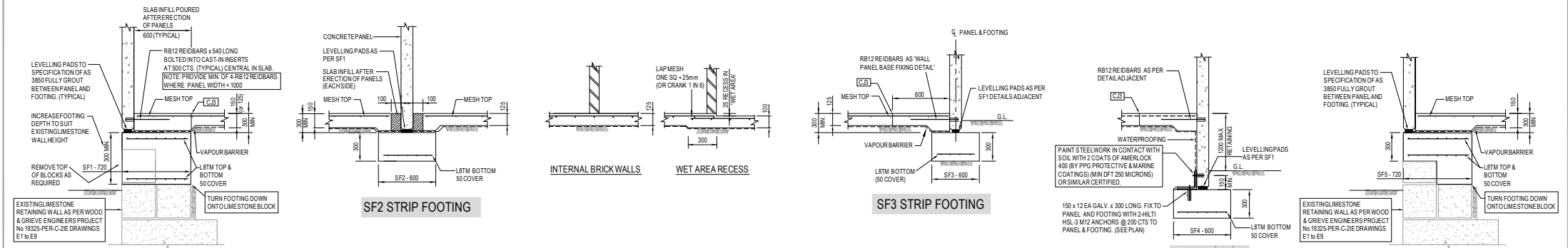
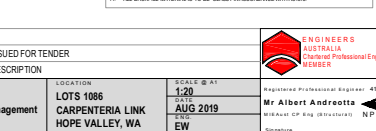
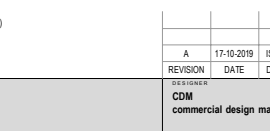
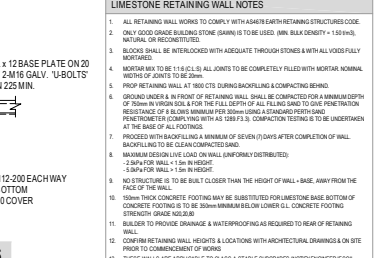
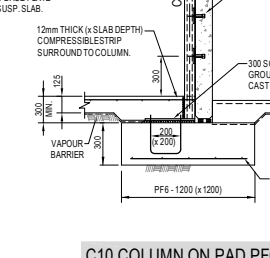
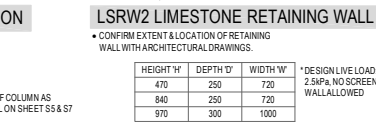
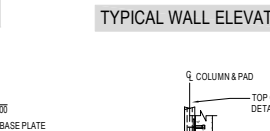
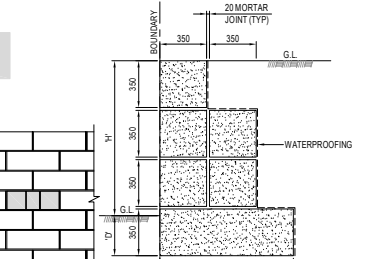
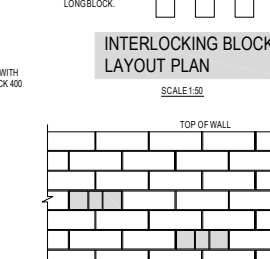
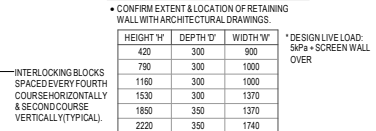
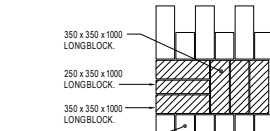
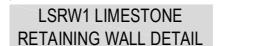
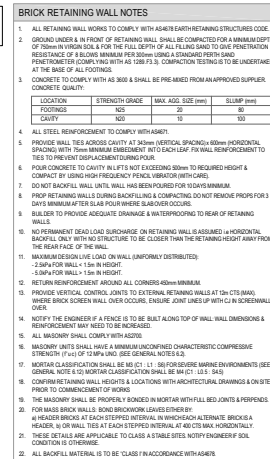
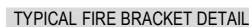
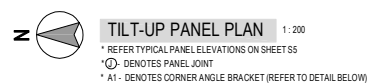


ANDREOTTA CARDENOSA CONSULTING ENGINEERS Suite 42, 11 Preston Street, C o m o , W . A . 6 1 5 2 Telephone: (08) 9367 6811 Email: acce@acce.net.au www.instagram.com/acce_au www.acce.net.au				ENGINEERS AUSTRALIA Chartered Professional Engineer MEMBER Registration Professional Engineer: 41952 Mr Albert Andreatta Member of Engineering Council of Australia Signature: _____ Date: 16 / 10 / 2019 Registered on the NRE in the Category of Structural National Professional Engineers Register			
DESIGNER	CDM commercial design management	LOCATION	LOTS 1086 CARPENTERIA LINK HOPE VALLEY, WA	SCALE	AS SHOWN	DATE	17-10-2019
PROJECT	PROPOSED OFFICE-WORKSHOP FACILITY	TITLE	FOOTING & GROUND SLAB DETAILS	SCALE	1:20	DATE	17-10-2019
REVISION	A	DATE	17-10-2019	DESCRIPTION	ISSUED FOR TENDER		

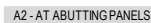




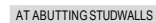
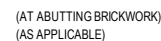
* LOCATE AS SHOWN ON SHEET S5 & S11 (DENOTED)



* LOCATE PLATES & BRACKETS NEAR TOP OF PANELS (BELOW ROOF SUPPORT MEMBERS / SUSPENDED SLAB) & AT MID-HEIGHT OF PANELS (WHERE HEIGHT OF PANEL IS GREATER THAN 5000)



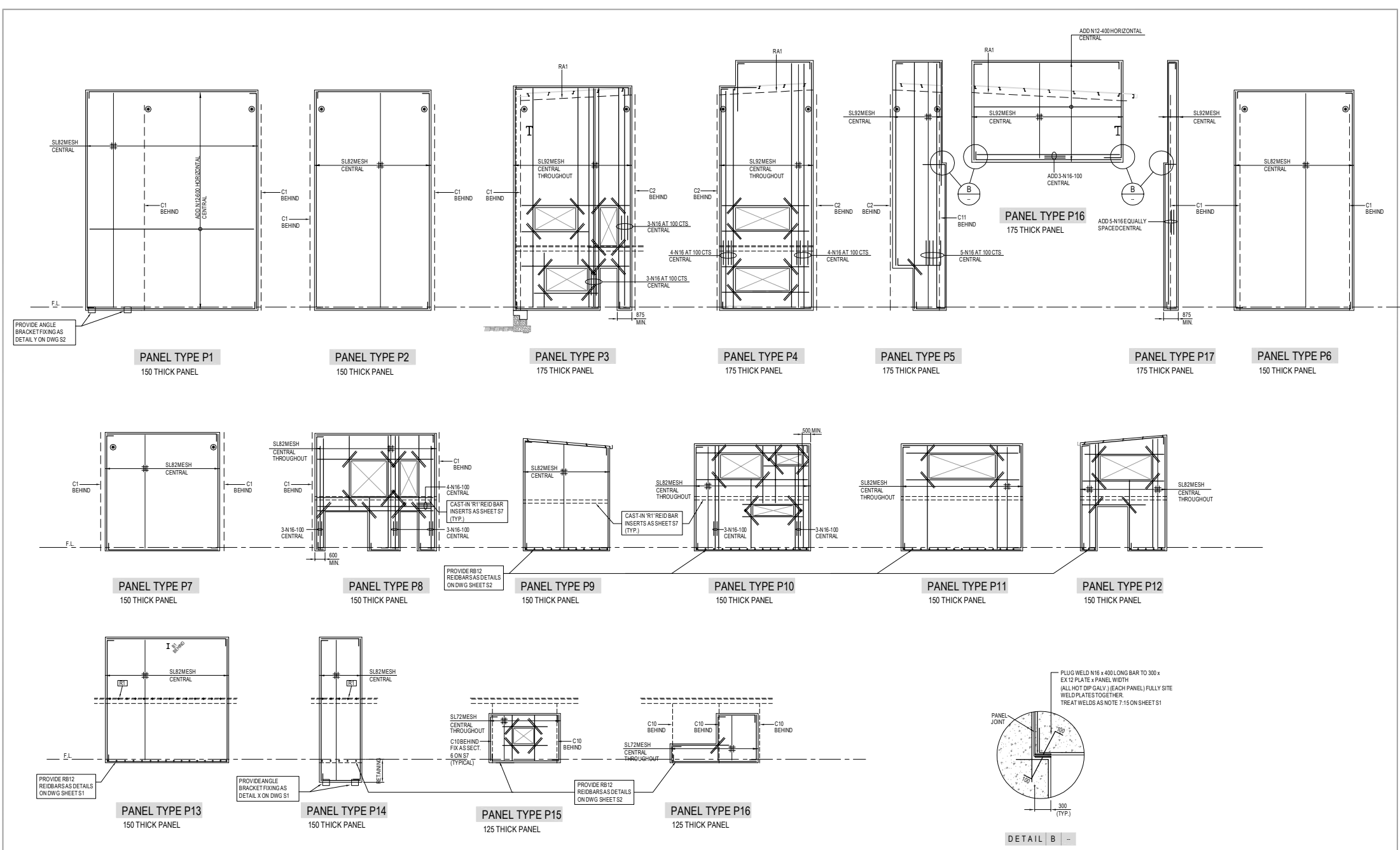
* LOCATE PLATES & BRACKETS NEAR TOP OF PANELS (BELOW ROOF SUPPORT MEMBERS / SUSPENDED SLAB) & AT MID-HEIGHT OF PANELS (WHERE HEIGHT OF PANEL IS GREATER THAN 5000)



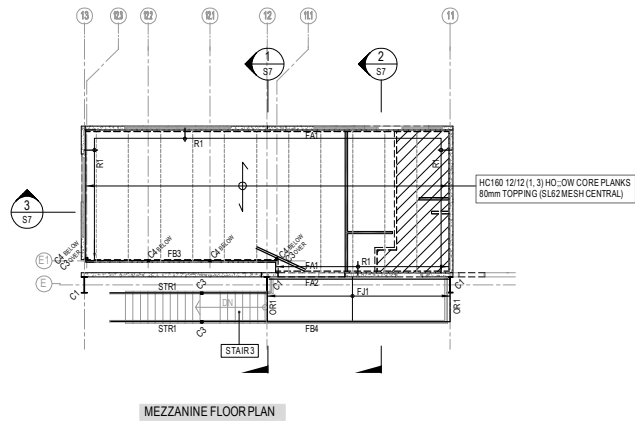
- FIX EX 10 PLATE BRACKETS TO COLUMNS AT 2500c/c MAX UNO (INCLUDING 200mm BELOW U/SIDE OF ROOF STEEL & 300mm ABOVE SLAB)

AC ANDREOTTA
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www.acce.net.au



CONCRETE PANEL ELEVATIONS
1:100 (AS VIEWED EXTERNALLY & INDICATIVE ONLY)
● DENOTES FIRE BRACKET AS DETAILS ON SHEET S4



OFFICE / MEZZANINE FLOOR PLANS

OFFICE FLOOR DESIGN LIVE LOAD: 3 KPa + STUD PARTITIONING

NOTES

- PROVIDE 2x12 x 1000 LONG BARS CENTRAL IN SLAB AT ALL RE-ENTRANT CORNERS, TYPICAL

LEGEND

- INDICATES CONCRETE FILLED CORES OF HOLLOW CORES
- CONCRETE PANEL WALLS
- WALLS OVER (STUD)
- NON-BEARING WALL BELOW
- INDICATES WET AREA: PROVIDE 25 RECESS IN TOPPING

COLUMNS

- C1 - 610 UB 113: REFER TO DETAIL ON SHEET S3
- C2 - 360 UB 45: REFER TO DETAIL ON SHEET S3
- PROVIDE 10mm LONG SLOTTED HOLES TO TOP FLANGING AS DETAIL 8 IN SHEET S3
- C3 - 75 x 5 SHS: REFER TO DETAIL ON SHEET S3
- C4 - 89 x 6 SHS: REFER TO DETAIL ON SHEET S3
- C5 - 125 x 6 SHS: REFER TO DETAIL ON SHEET S3
- C6 - 165 O.D. x 5.4 SHS: REFER TO DETAIL ON SHEET S3
- C7 - 310 UB 32: REFER TO DETAIL ON SHEET S3
- C8 - 100 x 5 SHS: REFER TO DETAIL ON SHEET S3
- C9 - 100 x 5 SHS: REFER TO DETAIL ON SHEET S3
- C10 - 125 x 75 x 6 SHS: REFER TO DETAIL ON SHEET S3
- C11 - 410 UB 54: REFER TO DETAIL ON SHEET S3
- PROVIDE 10mm LONG SLOTTED HOLES TO TOP FLANGING AS DETAIL 8 IN SHEET S3

FLOOR BEAMS

- FB1 - 440 UB 67
- FB2 - 290 PFC
- FB3 - 200 UB 22
- FB4 - 250 PFC
- FA1 - 150 x 12 EA: SEE DETAIL ALJACENT
- FA2 - 100 x 8 EA: FIX TO PANEL VIA WH1 HSB-300 ANCHORS AT 600 CTS MAX.

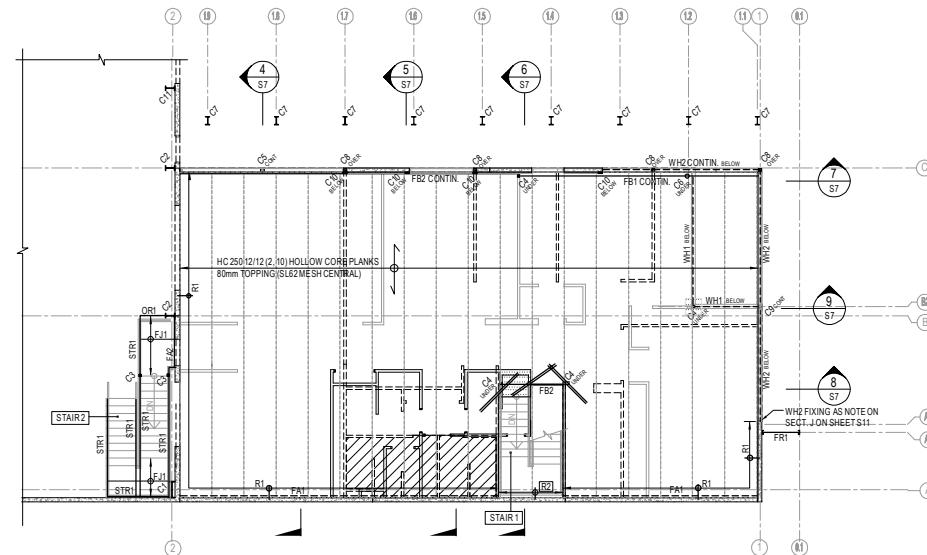
OTHERS

- WH1 - 150 PFC: EX 10 PLATE & 3/8" B&B BOLTS EACH END
- WH2 - 150 x 10 EA: EX 10 PLATE & 3/8" B&B BOLTS EACH END
- OR1 - 250 PFC
- STR1 - 200 PFC STRINGER: SEE DETAILS ON SHEET S8
- FJ1 - C150/12 JOISTS: AT 400 CTS MAX. EX 8 PLATE & 3/8" B&B BOLTS EACH END
- FR1 - C300/30 FULLY WELDED FRAME: - SEE SHEET S10
- R1 - R812 REID BARS: 140 LONG IN R812 T1 CAST IN INSETS AT 600 CTS MAX. CENTRAL IN TOPPING
- R2 - R812 REID BARS: 140 LONG IN R812 T1 CAST IN INSETS AT 200 CTS CENTRAL IN SLAB (TYP.)

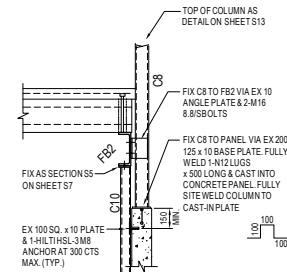
HOLLOW CORE FLOOR PLANKS NOTES:

- THE HOLLOW CORE PLANKS SHALL BE INSTALLED BY THE MANUFACTURER (DOLTA CORP LTD) OR A COMPETENT ERECTION CONTRACTOR. SET BEARING STRIPS WHERE REQUIRED & PROVIDE TEMPORARY SHORING AND BRACING AS NECESSARY TO MAINTAIN THE STABILITY OF THE STRUCTURE.
- TEMPORARY PROPPING OF PLANKS DURING CONSTRUCTION TO BE PROVIDED WHERE REQUIRED.
- PROVIDE TEMPORARY PROPPING AT 1m CTS BELOW ALL STEEL BEAMS SUPPORTING PLANKS. MAINTAIN PROPS 7 DAYS MIN. FOLLOWING PLACEMENT OF TOPPING.
- PROVIDE APPROVED FULL PACKING BELOW HOLLOW CORE PLANKS ABOVE ALL LOAD-BEARING BEAMS & WALLS.
- ALIGN AND GROUT THE KEYWAYS WITH A 1:1 SAND-CEMENT GROUT OR AN APPROVED TOPPING CONCRETE SO THAT THE KEYWAYS ARE PROPERLY FILLED. PROVIDE SUITABLE DAMS IN THE JOISTS AT PLANK ENDS AS REQUIRED.
- ATTACHMENTS AND FIXINGS TO THE HOLLOW CORE PLANKS SHALL BE IN ACCORDANCE WITH THE APPROVED DETAILS ONLY AND SHALL NOT IMPAIR OR REDUCE THE STRENGTH OF THE PLANKS.
- PENETRATIONS AND CHASES TO THE HOLLOW CORE PLANKS SHALL BE IN ACCORDANCE WITH THE APPROVED DETAILS ONLY AND SHALL BE APPROVED BY THE MANUFACTURER & THE STRUCTURAL ENGINEER.
- THE BUILDER SHALL PROVIDE AN IN-SITU CONCRETE TOPPING TO THE PLANKS AS NOTED BELOW. CLEAN DOWN THE DECK AND PREPARE THE PLANK SURFACE SO THAT THE CONCRETE IS MOIST BUT NOT OVERLY WET IMMEDIATELY PRIOR TO PLACING THE TOPPING. CURE AS NOTE 3.10 ON DWG S1.
- LAPS IN MESH TO BE MADE SO THAT THE OVERLAP MEASURED BETWEEN THE 2 OUTERMOST TRANSVERSE WIRES OF EACH SIDE OF EACH SHEET IS NOT LESS THAN THE SPACING OF THE WIRES PLUS 25mm.
- CONCRETE QUALITY:

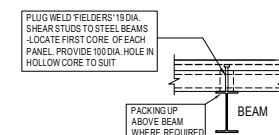
LOCATION	STRENGTH-GRADE	MAX. AGG. SIZE (mm)	SLOPP (mm)
CONC. TOPPING	N20 (B10.0)	10	80
- MANUFACTURER'S SHOP DETAIL DRAWINGS TO BE SUBMITTED TO THIS OFFICE FOR REVIEWING. (JOIST IS NOT INCLUDED IN ORIGINAL STRUCTURAL ENGINEERING FEE)
- PLANK MANUFACTURER IS TO ADD TOP STIFFENERS IN PLANKS AS NECESSARY TO AVOID JOGGING WITHIN PLANKS TO ENTER EVEN BEARING AT ALL SUPPORT POINTS.
- BUILDER IS TO ALLOW FOR CAMBER IN FLOOR PLANKS WHEN FABRICATING COLUMNS OVER.



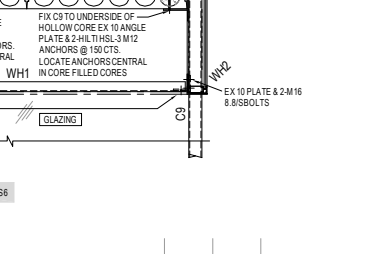
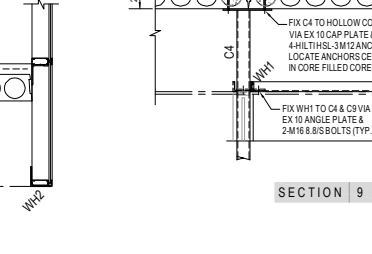
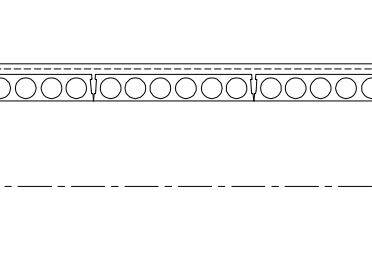
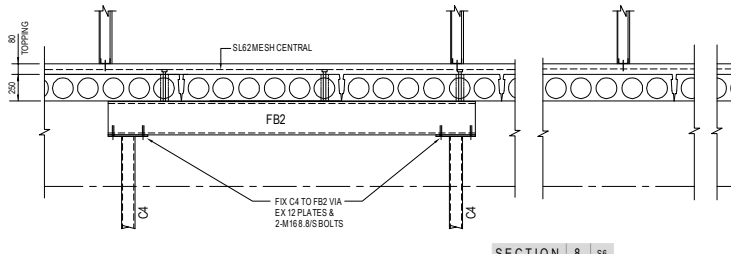
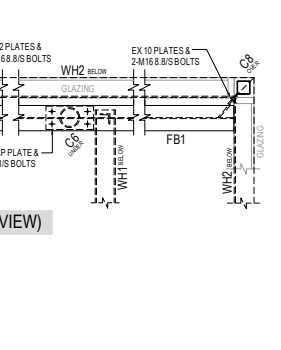
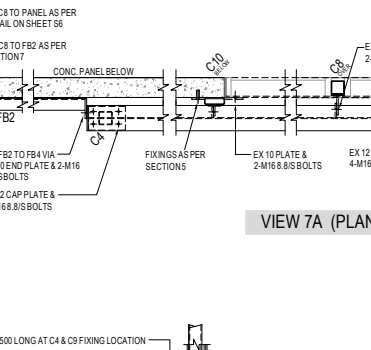
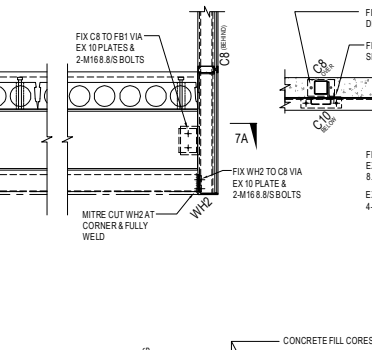
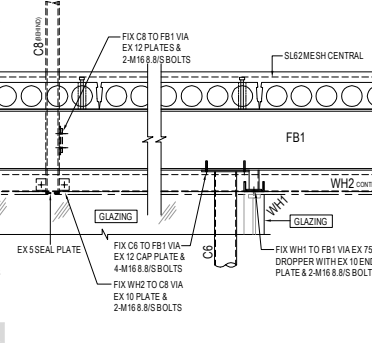
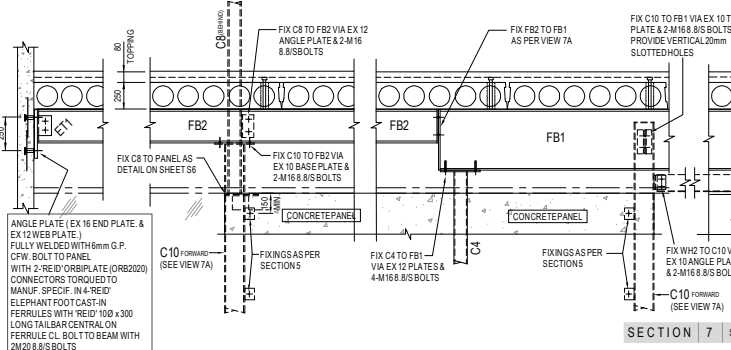
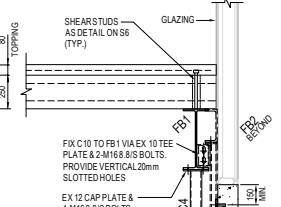
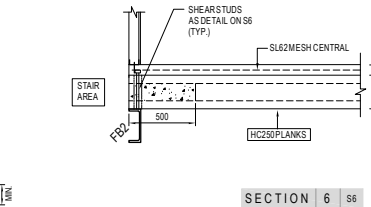
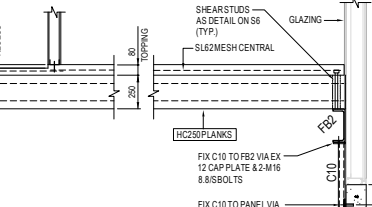
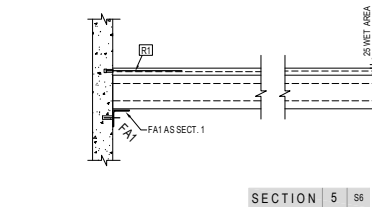
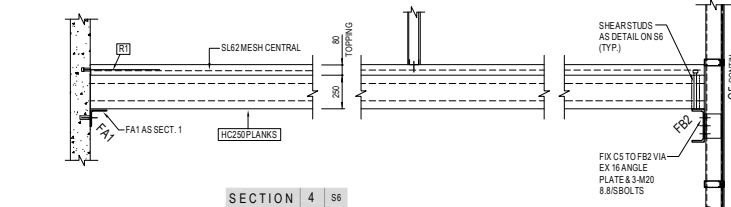
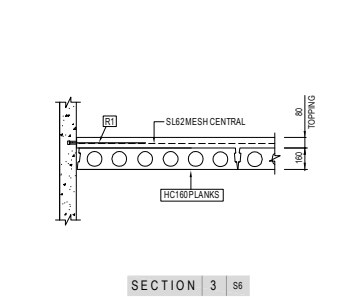
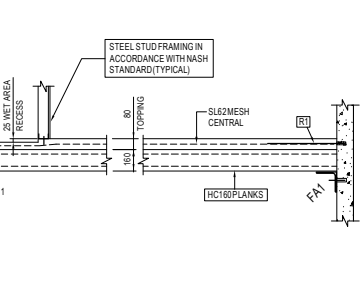
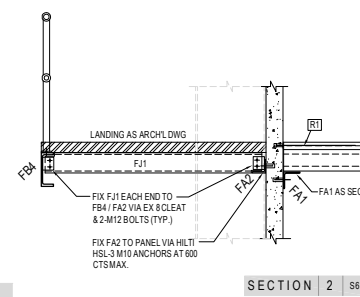
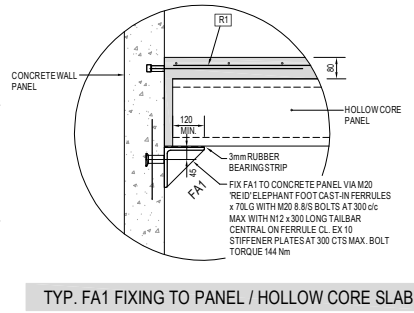
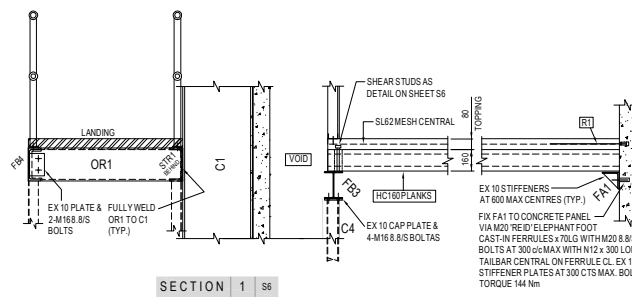
OFFICE FLOOR PLAN

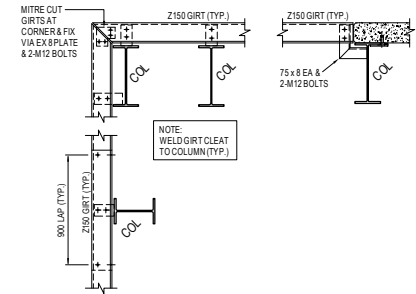
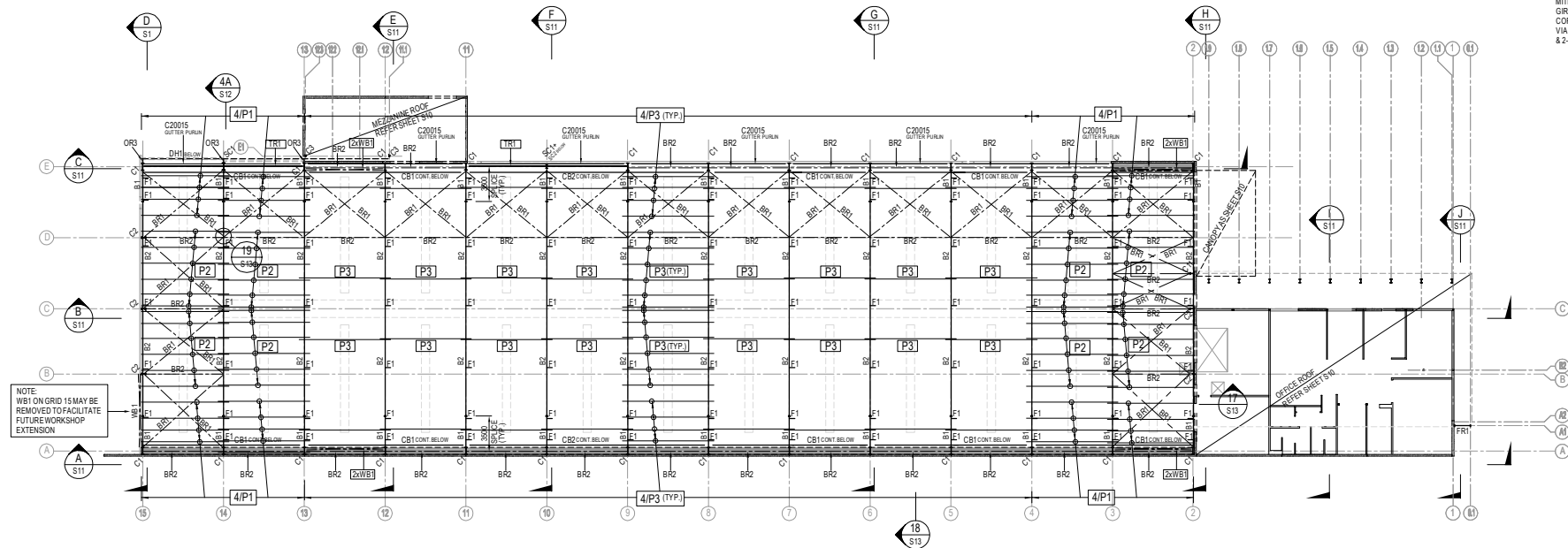


C8 COLUMN ON PANEL / HOLLOW CORE



TYP. BEAM / HOLLOW CORE SLAB FIXING





GIRT / COLUMN DETAILS (PLAN VIEW)

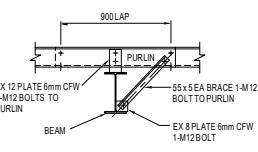


ROOF STEEL PLAN

ROOF PLAN NOTES	
• MINIMUM STEEL TO STEEL CONNECTION EX 10 PLATE & 2-M16 8.8/S BOLTS OR 6mm GP CONTINUOUS FILLET WELD (UNO).	
• ALL BOLTING TO BE CATEGORY 8.8/UNO	
• PROVIDE ADDITIONAL PURLINS & GIRTS AS REQ'D TO SUIT SHEETING SUPPORTS, CATEGORY, OPENINGS ETC. STOOL UP PURLINS TO SUIT GUTTERS	
• PURLINS & GIRTS SHOWN ON PLANS ARE INDICATIVE ONLY. BUILDER / SHOP DETAILER TO DETERMINE EXACT NUMBER OF PURLINS / GIRTS	
LEGEND	
	INDICATES BOX GUTTER
COLUMNS	
C1 - 610 UB 113: REFER TO DETAIL ON SHEET S3	
C2 - 300 UB 45: REFER TO DETAIL ON SHEET S3	
PROVIDE 75mm LONG SLOTTED HOLES TO TOP FIXING AS DETAIL B IN SHEET S10	
C3 - 75 x 5 SHS	
C11 - 410 UB 54: REFER TO DETAIL ON SHEET S3	
PROVIDE 75mm LONG SLOTTED HOLES TO TOP FIXING AS DETAIL B IN SHEET S10	
SC1 - 610 UB 113	
SC2 - 300 UB 45	
BEAMS	
B1 - 610 UB 113	
B2 - 410 UB 54	
DH1 - 310 UB 32	
CRANERUNWAY BEAMS:	
CB1 - 410 UB 54 + 380 PFC (SPAN CONTINUOUS)	END NOT
CB2 - 410 UB 54 + 380 PFC (SPAN CONTINUOUS)	END NOT
NOTE:	
FINAL CRANE LOADINGS ON STRUCTURE TO BE PROVIDED TO THIS OFFICE PRIOR TO ISSUED FOR CONSTRUCTION TO VERIFY DESIGN	

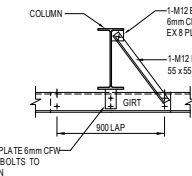
MAXIMUM SOLAR PANEL WEIGHT = 12 kg/m²

TRUSSES / FRAME	
TR1 - 3800 DEEP OVERALL (REFER ELEVATION ADJACENT)	
FR1 - FRAME AS PER SHEET S10	
OTHER	
ET1 - 150 PFC	
RA1 - 100 x 8 EA	
OR3 - 150 x 6 SHS OUTRIGGER	
BRACINGS	
BR1 - 65 x 5 EA: EX 10 PLATE & 2-M16 8.8/S BOLTS EACH END. R10 SAG ROD TO PURLIN OVER NEAREST MIDSPAN	
BR2 - 100 x 5 SHS: EX 10 TEE PLATE & 2-M16 8.8/S BOLTS EACH END.	
WB1 - 75 x 6 EA: EX 10 PLATE & 2-M16 8.8/S BOLTS EACH END.	
WB2 - 100 x 5 SHS: EX 10 TEE PLATE & 2-M16 8.8/S BOLTS EACH END.	
F1, F2, F3 - FLY BRACES AS DETAILS ADJACENT. REFER TO SECTIONS ON S11 FOR LOCATIONS	
PURLINS / GIRTS	
P1 - Z150/19 AT 1000 CTS MAX. LAP 900 AT INTERNAL SUPPORTS BRIDGING AT THIRD POINTS	
P2 - Z150/19 AT 1500 CTS MAX. LAP 900 AT INTERNAL SUPPORTS BRIDGING AT THIRD POINTS	
P3 - Z150/12 AT 1000 CTS MAX. LAP 900 AT INTERNAL SUPPORTS BRIDGING AT THIRD POINTS	
P4 - Z150/12 AT 1500 CTS MAX. LAP 900 AT INTERNAL SUPPORTS BRIDGING AT THIRD POINTS	
G1 - Z150/12 AT 1800 CTS MAX. LAP 900 AT INTERNAL SUPPORTS BRIDGING MIDSPAN	



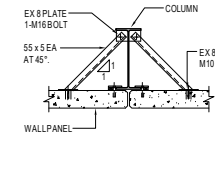
F1 TYPICAL FLYBRACE

* DENOTED F1 ON ROOF STEEL PLAN



F2 TYP. COLUMN FLYBRACE

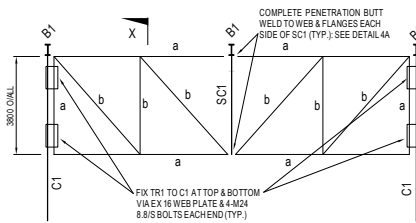
(AT GIRTS)
* REFER TO SECTIONS C & D FOR LOCATIONS



F3 TYP. COLUMN FLYBRACE

(AT WALL PANEL)
* REFER TO SECTIONS A & H FOR LOCATIONS

VIEW X-X



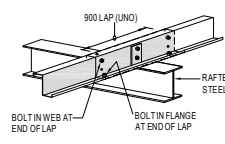
TR1 TRUSS ELEVATION

SCALE 1:100

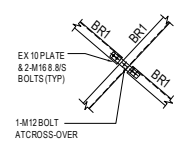
* ALL TRUSS MEMBERS TO BE FULLY WELDED
* ALL VERTICAL MEMBERS TO BE SPACED AT EQUAL CENTRES BETWEEN SUPPORT MEMBERS
* TREAT STEEL WORK AS NOTE 7.14 ON STRUCTURAL DRAWING S1

MEMBERS SCHEDULE

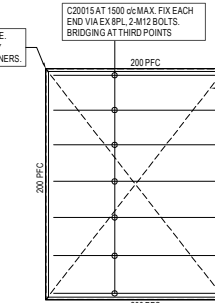
a - 410 UB 60
b - 150 x 6 SHS



TYPICAL PURLIN LAP DETAIL



BR1 BRACING AT CROSS-OVER



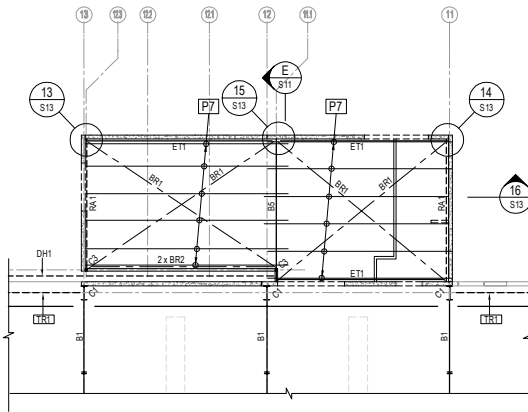
TYPICAL SLIDING DOOR PANEL ELEVATION

SCALE 1:100
* PROVIDE SLIDING DOOR ROLLERS & FIXING BRACKETS AS ARCH. DWG
* REFER TO ARCH. DWG'S FOR DIMENSIONS & FIXINGS.

50 x 3 FLAT PLATE WELDED TO HORIZONTAL & VERTICAL MEMBERS - FULLY SITE WELD SIDS TO PERIMETER FRAME

EX 8 PLATE & 2-M12 BOLTS

VIEW Y-Y



MEZZANINE ROOF PLAN

MAXIMUM SOLAR PANEL WEIGHT = 12 kg/m²

ROOF STEEL PLANS - PART 2

ROOF PLAN NOTES

- MINIMUM STEEL TO STEEL CONNECTION EX 10 PLATE & 2-M16 8.8 BOLTS OR 6mm GP CONTINUOUS FILLET WELD (UNO).
- ALL BOLTING TO BE CATEGORY 8.8S UNO.
- PROVIDE ADDITIONAL PURLINS AS REQ'D TO SUIT SHEETING SUPPORTS AT ENDS, OPENINGS ETC. STUOL UP PURLINS TO SUIT GUTTERS.
- PURLINS SHOWN ON PLAN ARE INDICATIVE ONLY. BUILDER/SHOP DETAILER TO DETERMINE EXACT NUMBER OF PURLINS.

LEGEND

INDICATES BOX GUTTER

COLUMNS

C1- 610 UB 113
C2- 360 UB 45
C3- 75 x 5 SHS
C7- 310 UB 32
C5- 125 x 6 SHS
C8- 100 x 5 SHS
C9- 100 x 5 SHS
C11- 410 UB 54
C13- 75 x 5 SHS STUB

ROOF BEAMS

B2- 410 UB 54
B3- 360 UB 51
B4- 150 PFC
B5- 200 UB 18
B6- C200x4 + C200x4 BOLT TOGETHER VIA M10 BOLTS AT 500 CTS MAX.
B7- 300 PFC
B8- 125 PFC
B9- 100 PFC AT 600 CTS MAX.

OTHERS

OR2- 150 UC 23 FULLY WELD TO COLUMN
ET1- 150 PFC
RA1- 100 x 8 EA
WH1- 150 PFC
EX 10 PLATE & 2-M16 8.8 BOLTS EACH END
FR1- FULLY WELDED FRAME AS DETAIL ADJACENT
DH1- 310 UB 32
ST1- 60 x 60 x 4.5 CHS. EX 10 TEE PLATE & 2-M16 8.8 BOLTS EACH END
VO1- 75 x 5 EA. EX 10 TEE PLATE & 2-M16 8.8 BOLTS EACH END
EX 10 PLATE & 2-M16 8.8 BOLTS TO EACH GIRT

PURLINS / GIRTS / CEILING JOISTS

P5- Z15012 PURLINS AT 1200 CTS MAX.
1000 MAX END SPACING
LAP 900 AT INTERNAL SUPPORTS
BRIDGING AT MIDSPAN

P6- C20015 PURLINS AT 1200 CTS MAX.
1000 MAX END SPACING
BRIDGING AT MIDSPAN

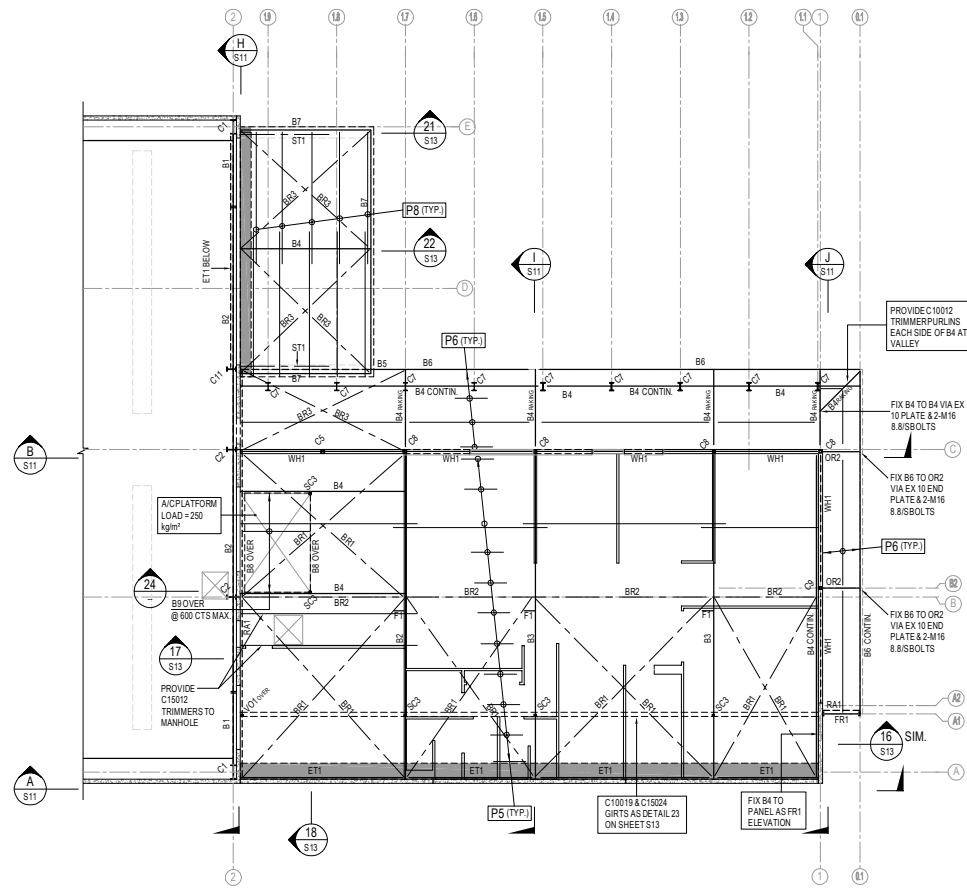
P7- Z15019 PURLINS AT 1200 CTS MAX.
1000 MAX END SPACING
LAP 900 AT INTERNAL SUPPORTS
BRIDGING AT THIRD POINTS

P8- C10019 PURLINS AT 1200 CTS MAX.
1000 MAX END SPACING
LAP 900 AT INTERNAL SUPPORTS
BRIDGING AT MIDSPAN

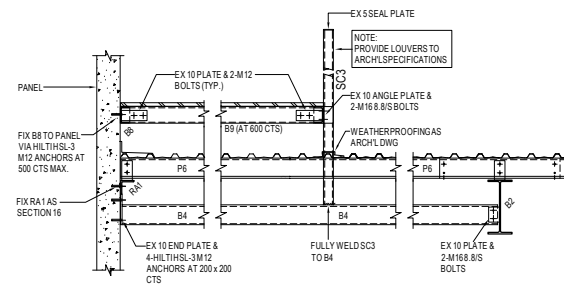
BRACINGS

BR1- 69x5 EA. EX 10 PLATE & 2-M16 8.8 BOLTS EACH END. R10 SAG ROD TO PURLIN OVER AT MIDSPAN.
BR2- 100 x 5 SHS. EX 10 TEE PLATE & 2-M16 8.8 BOLTS EACH END.
BR3- 30 x 0.8mm BMT GALT. STRAP FIX EACH END VIA 2 SERIES 500 TKS SCREWS & TKS SCREW TO UNDERSIDE OF EACH PURLIN. FULLY TENSION PRIOR TO FIXING.

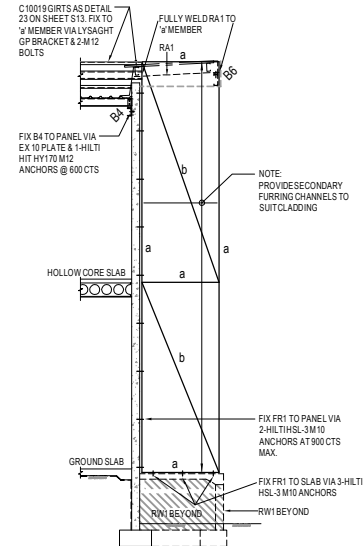
F1- FLYBRACE. SEE SHEET 03



OFFICE ROOF PLAN



SECTION 24



FR1 FRAME ELEVATION 1:50

- * REFER TO ARCH. DWGS FOR DIMENSIONS
- * ALL FRAME MEMBERS TO BE FULLY WELDED
- * ALL MEMBERS TO BE SPACED EQUALLY
- * TREAT STEEL WORK AS NOTE 7:14 ON STRUCTURAL DRAWING S1

FRAME MEMBERS

a- C30030
b- C20024

